Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Interactive AI Systems for Intuitive Human-Robot Collaboration in Home and Office Environments			
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ),** **Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (✓),** **Next-gen Communication ( ), Advanced Robotics & Manufacturing (✓), Quantum ( )**			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Robot Control and Intelligence Technologies	*Robotics / Automation Machinery*	Robotics / Automation Machinery Software
Concept of Proposed Technology	This proposal aims to develop **an interactive AI system that enables robots to collaborate naturally with human partners in home and office environments**. Unlike conventional robot learning approaches that rely primarily on static, offline demonstrations, *this technology focuses on robots that can incorporate and learn from real-time human interaction during task execution*. The system allows human collaborators to guide, correct, and teach robots using natural communication modalities such as spoken language, gestures, and "short video" demonstrations			

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	(e.g., “watch this and repeat”). Through repeated deployment, interaction, and retraining cycles, the robot continuously improves its task performance, autonomy, and adaptability to new environments. By enabling robots to learn directly from human feedback during everyday operation, the proposed technology provides a practical foundation for robust, human-centered robotic systems that can be effectively deployed — **and that can continuously improve** — in real-world settings. Key Features ● **Interactive, Human-in-the-Loop Robot Learning:** Robots learn and adapt during task execution through real-time human guidance, corrections, and demonstrations, enabling natural forms of collaboration with real people. ● **Continuous Improvement via Deployment and Retraining:** The learned AI policies are iteratively improved through repeated deployment and retraining cycles, leading to increased autonomy, improved performance, and more efficient adaptation to new environments. ● **Real-World Evaluation in Home/Office Settings:** The system is evaluated through long-term deployments in realistic home and office environments, enabling validation of robustness, usability, and collaborative performance in everyday conditions.
Rationale for Joint R&D Project	While the research institute contributes expertise in interactive robot learning, human-robot interaction, and adaptive AI algorithms, **industrial partners provide essential capabilities in system integration, hardware platforms, and access to realistic home and office environments**. Joint R&D enables iterative deployment, real-world data collection, and repeated system refinement under practical conditions that cannot be fully replicated in laboratory settings. By directly addressing challenges related to human interaction, safety, and usability in real environments, this technology establishes a strong foundation for assessing the real-world viability and commercialization potential of collaborative robotic systems. Through such collaboration, the project accelerates the transition from interactive AI research to deployable robotic systems that are robust, usable, and well-suited for real-world human-robot collaboration.
Key Words	1. Interactive Human-Robot Collaboration 2. Human-in-the-Loop Robot Learning 3. Adaptive and Interactive AI Systems 4. Real-World Robot Deployment 5. Service, Household, and Collaborative Robots

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	This project will develop and validate an interactive AI system that enables intuitive and effective human-robot collaboration in real-world home and office environments. The specific objectives include: • **Developing interactive human-in-the-loop AI methods for robot learning and control.** These methods will allow robots to receive, interpret, and respond to real-time human guidance, corrections, and demonstrations during task execution. • **Building a scalable system for iterative deployment, data collection, and retraining**. This will enable continuous improvement of robot performance and adaptability across multiple deployments and environments. • **Integrating and validating the interactive AI system on robotic platforms in realistic home and office settings**. This demonstrates robust operation, safe human-robot interaction, and reliable task execution under real-world conditions. • **Evaluating performance improvements across *repeated* deployments**. This includes showing increased robot autonomy (better task performance), reduced human intervention, and more data-efficient adaptation to new environments.			
R&D Target Specifications	The target performance metrics of this project are as follows: • **Interactive Responsiveness (Ability to Respond to Human Input)** Achieve reliable real-time robot response to human guidance, corrections, and demonstrations during task execution, with **sub-second response latency** and correct execution in **≥ 90% of interaction events** across speech, gestures, and other modalities. • **Autonomy Improvement *across* Deployments** Demonstrate increasing robot autonomy through repeated deployment and retraining cycles, achieving **70–80% autonomous task completion** for supported tasks and *a decreasing need for human intervention over time*. • **Efficient Adaptation to New Environments** Reduce the amount of demonstration or human intervention data required to adapt to new home and office environments by **at least 30%** across successive deployments. • **Robust Real-World Operation (Task Success Rate)** Achieve stable and safe operation in realistic home and office environments, with **≥ 80% task success rate** after at least 5 deployment cycles for tasks under everyday operating conditions.			

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R&D Plan	The proposed project will be conducted jointly over three years, following a staged development and validation strategy that progresses from interactive AI system development to real-world deployment and evaluation. **Year 1: Interactive AI System Development and Simulation-Based Validation** ● Develop core interactive AI methods for human-in-the-loop robot learning, including real-time interpretation of human guidance, corrections, and demonstrations. ● Establish a simulation-based evaluation environment to design and validate interaction workflows, data collection pipelines, and iterative retraining procedures. ● Conduct initial validation of interactive responsiveness and autonomy improvement through controlled simulation experiments. **Year 2: System Integration and Pilot Deployment in Realistic Environments** ● Integrate the interactive AI system on robotic platforms in laboratory-scale and pilot home/office environments. ● Customize and optimize the system for real hardware constraints, sensor noise, and real-time operation. ● Conduct pilot deployments to collect real-world interaction and correction data, AND ● Perform iterative retraining to demonstrate increased autonomy and reduced human intervention. **Year 3: Multi-Site Deployment and Robustness Evaluation** ● Expand deployment to multiple home and office environments to evaluate scalability, robustness, and adaptability. ● Perform repeated deployment and retraining cycles to demonstrate data-efficient adaptation to new environments. Verify stable performance under diverse real-world conditions. ● Evaluate system reliability, safety, and usability to assess readiness for pilot-scale deployment and future commercialization.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the development of the core interactive AI technologies and system-level intelligence required for intuitive human-robot collaboration. The institute will lead research on interactive robot learning, human-robot interaction, and adaptive AI methods that enable robots to learn from real-time human guidance and feedback. The specific roles and responsibilities of the research institute include: ● Development of interactive human-in-the-loop AI methods that enable robots to interpret and respond to human guidance, corrections, and demonstrations during task execution. ● Design of learning and retraining workflows for continuous improvement of robot autonomy across repeated deployments. Includes developing a platform for collecting and managing human interaction data at scale. ● Development of simulation and real-world “mock” environments to serve as evaluation platforms to support algorithm development, testing, and performance analysis. ● Close collaboration with the Korean SME partner to support system integration, algorithm tuning, data collection, and real-world evaluation strategies.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the integration and real-world validation of the robotic systems required for the proposed interactive AI technology. The SME will play a central role in translating the interactive AI methods into practical, deployable robotic systems suitable for home and office environments. The specific roles and responsibilities of the Korean SME partner include: ● Provision and integration of robotic hardware platforms, including mobile or fixed robot manipulation systems, sensors, and supporting system components. ● System-level integration of interactive AI software with robotic hardware, sensing, and control systems. ● Support for pilot deployment in realistic home and office environments, including execution of real-world trials and data collection. ● Contribution to system refinement, reliability improvement, and safety validation based on deployment experience and practical operating constraints. ● Leadership in evaluating commercialization potential and future product development pathways leveraging industrial expertise.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint project aims to produce a pilot-scale interactive AI system that enables intuitive and effective human-robot collaboration in home and office environments. The project will demonstrate that robots can learn from real-time human guidance (via modalities such as speech and gestures) and continuously improve their autonomy and performance through repeated deployment and retraining cycles. The project will provide the Korean SME partner with core technologies and experience for deploying and continually improving interactive, human-centered robotic systems in real-world environments. The outcomes will support the development of safer, more usable, and more adaptable collaborative robots, and establish a strong foundation for future product development and commercialization, including opportunities to evaluate the proposed technology with actual users in meaningful commercial settings. More broadly, the project will contribute to advancing safe and reliable human-centered AI and robotics technologies with real-world impact.

6. References

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(2) Grannen, Karamcheti, Wulfe, Sadigh, "ProVox: Personalization and Proactive Planning for Situated Human-Robot Collaboration." In *IEEE Robotics and Automation — Letter (RAL), vol. 10, no. 8, pp. 8451-8458, Aug. 2025.* DOI: 10.1109/LRA.2025.3583460

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(5) Karamcheti, Nair, Chen, Kollar, Finn, Sadigh, Liang, "Language-Driven Representation Learning for Robotics." In *Robotics: Science and Systems (RSS), Jul. 2023.* https://arxiv.org/abs/2302.12766.

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☐ **Terms:**

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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